Spark Homes Repair Estimator - Submission Writeup

A mobile-first, offline-ready field estimator built as a static PWA with vanilla HTML, CSS, and JavaScript. It stores projects locally, supports configurable rooms, captures photos, and exports either Excel-only or ZIP-with-photos depending on the walkthrough.

1. Most interesting design / UX decision	I designed the app around a command-center walkthrough flow: persistent running total at the top, room tabs as the primary navigation, group-level progress, and large thumb-friendly controls. The goal was to make the estimate usable while standing in a house, not just as a desktop spreadsheet replacement. I also added a Missed-Cost Sentinel so the app gently flags likely omissions such as unreviewed systems, $0 bathrooms, or old HVAC equipment before export.
2. What is broken or fragile	The build is intentionally offline/local only, so very large photo sets can hit browser localStorage limits. Serial-number support is best-effort: it detects serial-like patterns from image filenames and supports photo evidence, but it does not run full OCR on image pixels. Voice input depends on browser speech-recognition support, so it is a helpful bonus rather than a required workflow.
3. Creative addition	The main creative additions are the Deal Analyzer, Aging Penalty Engine, Voice Walkthrough, and Missed-Cost Sentinel. The Deal Analyzer turns the repair estimate into an acquisition decision by calculating projected resale margin, holding/financing costs, and estimated profit. The Aging Penalty Engine highlights old furnace, condenser, and water-heater equipment so big-ticket system misses are harder to overlook.
4. What I would ship next with two more days	I would add true on-device OCR for serial plates, a safer photo-storage layer using IndexedDB instead of localStorage, and a share sheet flow optimized for sending the exported file directly from iOS/Android. I would also add a compact QA dashboard comparing this estimate against past projects by room and cost category.
AI tools used	AI assisted with code review, feature-gap analysis, refactoring guidance, and final packaging/QA. I made the product decisions, checked the contest requirements against the implementation, and verified the final static files before packaging.

Export behavior: If the project has no photos, Export downloads a standalone Excel workbook. If photos exist, Export downloads a ZIP containing the Excel workbook plus a photos folder. The service worker and manifest are included for installable offline use on Android and iOS.